

OVIEDO, Spain

WMO: 080150

Lat: 43.3533N

Lon: 5.8742W

Elev: 339

StdP: 97.32

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.2	(c) 1.3	(d) -2.9	(e) 3.1	(f) 2.9	(g) -1.5	(h) 3.5	(i) 4.0	(j) 10.4	(k) 11.8	(l) 9.2	(m) 11.0	(n) 1.9	(o) 170	(p) 0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 7.5	(c) 27.3	(d) 20.4	(e) 25.5	(f) 19.5	(g) 24.0	(h) 18.8	(i) 21.2	(j) 25.9	(k) 20.2	(l) 24.2	(m) 19.5	(n) 23.0	(o) 2.6	(p) 60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.9	23.4	18.7	14.1	22.1	18.1	13.6	21.3	63.0	25.7	59.5	24.3	56.8	23.3	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.9	6.2	5.0	-1.7	31.6	1.2	2.3	-2.5	33.3	-3.3	34.6	-3.9	35.9	-4.8	37.6
			-2.3	23.7	1.3	1.5	-3.3	24.8	-4.0	25.7	-4.7	26.5	-5.7	27.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	8.5	8.7	10.5	11.5	14.0	16.9	18.7	19.1	17.5	15.1	10.6	8.9		
	DBStd	4.84	3.28	3.31	3.38	3.03	2.97	2.75	2.21	2.22	2.46	2.98	3.03	3.30		
	HDD10.0	272	69	58	34	18	4	0	0	0	0	2	27	60		
	HDD18.3	1933	306	271	242	205	138	59	22	17	45	106	231	291		
	CDD10.0	1501	22	21	51	63	128	207	269	282	224	161	46	27		
	CDD18.3	119	0	0	1	1	3	16	33	40	19	6	0	0		
	CDH23.3	508	0	0	1	8	25	98	124	156	74	23	1	0		
	CDH26.7	97	0	0	0	1	3	26	18	34	11	3	0	0		
(14) Wind	WSAvg	2.0	2.3	2.3	2.3	2.1	1.9	1.8	1.7	1.7	1.8	1.8	2.1	2.1		
(15) Precipitation	PrecAvg	1033	102	93	88	102	82	65	48	57	59	96	135	90		
	PrecMax	1353	212	233	193	255	224	222	135	185	107	210	225	179		
	PrecMin	793	16	15	5	3	26	16	9	16	21	17	24	5		
	PrecStd	161	54	59	53	62	44	47	30	40	26	50	52	55		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.9	19.7	22.4	24.1	26.2	30.3	28.8	30.1	28.1	25.9	20.7	18.6		
		MCWB	13.4	12.5	13.8	16.1	18.6	21.3	21.7	22.0	19.9	17.8	14.9	12.2		
	2%	DB	16.7	17.1	19.7	20.6	23.0	25.8	26.2	26.8	25.2	23.2	18.5	16.8		
		MCWB	11.9	11.6	13.2	14.5	17.3	19.5	20.5	20.6	19.5	17.1	14.1	11.7		
	5%	DB	15.1	15.3	17.8	18.6	20.9	23.5	24.5	24.7	23.4	21.3	16.7	15.4		
		MCWB	10.8	10.6	12.2	13.5	16.1	18.3	19.5	19.6	18.5	16.3	13.2	11.1		
	10%	DB	13.5	13.7	15.9	16.6	18.9	21.6	22.9	23.2	21.8	19.7	15.1	14.0		
		MCWB	9.9	9.8	11.2	12.5	15.0	17.4	18.7	18.8	17.8	15.6	12.1	10.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.2	14.0	15.0	17.1	19.4	21.8	22.5	23.1	21.2	19.4	16.2	13.5		
		MCDB	17.8	17.6	20.1	22.9	24.4	28.6	27.5	28.9	25.9	23.2	19.2	16.1		
	2%	WB	12.6	12.2	13.7	15.1	17.7	20.1	20.8	21.3	20.0	18.0	14.7	12.4		
		MCDB	15.9	15.8	18.6	19.6	22.5	25.2	25.1	25.6	24.2	21.5	17.3	15.3		
	5%	WB	11.4	11.1	12.6	14.0	16.5	18.9	19.9	20.2	19.0	17.0	13.7	11.7		
		MCDB	14.2	14.3	16.8	17.7	20.2	22.8	23.6	23.6	22.6	20.0	15.9	14.6		
	10%	WB	10.3	10.3	11.7	12.9	15.4	17.8	19.1	19.3	18.2	16.2	12.6	10.8		
		MCDB	12.8	13.0	15.4	16.1	18.4	20.9	22.2	22.3	21.0	18.9	14.7	13.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	6.7	7.6	7.7	7.5	7.3	7.2	7.5	7.7	7.3	5.9	6.1		
		MCDBR	7.9	9.3	10.6	10.9	10.7	10.9	10.0	10.5	10.3	9.6	8.3	7.5		
		MCWBR	5.2	5.8	5.9	6.0	5.8	5.7	5.0	5.1	5.3	5.2	5.4	4.8		
		MCDBR	7.1	8.2	9.6	10.0	9.9	9.9	9.2	9.4	9.4	8.5	7.4	6.6		
	5% WB	MCWBR	5.0	5.5	5.8	5.8	5.7	5.5	4.9	4.9	5.1	5.0	5.1	4.6		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.324	0.359	0.377	0.391	0.399	0.388	0.385	0.368	0.350	0.325	0.309		
	taud		2.514	2.473	2.383	2.342	2.329	2.328	2.361	2.380	2.424	2.483	2.513	2.522		
	Ebn at Noon		824	875	880	888	882	872	878	870	859	832	801	795		
	Edh at Noon		70	86	106	119	124	125	120	114	101	84	70	64		
(44) All-Sky Solar Radiation	RadAvg		1.47	2.29	3.45	4.57	5.41	5.84	5.96	5.39	4.31	2.77	1.66	1.34		
	RadStd		0.16	0.32	0.45	0.46	0.44	0.44	0.34	0.35	0.17	0.26	0.20	0.17		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page